Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

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D6.3 Mechanisms to promote fairness.

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Authors: Natalia Calvo Barajas, Ginevra Castellano

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Project Members:

Max-Planck-Gesellschaft Zur Forderung Der Wissenschaften EV (MPI-SWS)

Technische Universiteit Eindhoven (TUE)

Kungliga Tekniska Hoegskolan (KTH)

Uppsala Universitet (UU)

Stichting Koninklijk Nederlands Lucht - En Ruimtevaartcentrum (NLR)

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List of Abbreviations

ANT Actor-Network Theory

HAI Human-Agent Interaction

HRI Human-Robot Interaction

MAS Multi-agent System

WP Work Package

Introduction

This report is the deliverable for Task 6.3 in the third year of the SymAware project. It contributes to Work Package (WP) 6, *Trustworthy and ethical human-agent interaction*, and specifically addresses the objectives of Task 6.3, which focuses on investigating mechanisms to promote fairness and prevent bias in the computational methods and interaction mechanisms developed in the project. The deliverable is grounded in the observation that situational awareness in multi-agent systems (MASs) is not only a matter of sensing, modelling, and coordination, but also of whether the resulting interaction is perceived as trustworthy and works equitably across different users and contexts. If fairness is not considered explicitly, autonomy and learning-based components may introduce systematic disparities in how information is provided, how support is allocated, or how decisions affect different user groups, thereby undermining trust, acceptance, and ultimately the quality of situational awareness.

Situational awareness is commonly described in three levels, namely the perception of relevant elements in the environment, the comprehension of their meaning, and the projection of their future status [1]. In Human-Agent Interaction (HAI) and MASs, fairness is relevant at all three levels. At the level of perception, fairness concerns whether relevant cues and information are equally available and salient for different users. At the level of comprehension, fairness concerns whether explanations, interface representations, and transparency cues are equally understandable and usable. At the level of projection, fairness concerns whether users can anticipate agent behaviour, limitations, and uncertainty with comparable confidence. As such, fairness becomes a fundamental ethical principle and a requirement for trustworthy situational awareness [2].

Trust and acceptance in automation therefore depend on *appropriate reliance* rather than maximal reliance. Fairness and reliance are closely intertwined, because users' willingness to follow or question automated support depends not only on objective system performance, but also on how fair, interpretable, and trustworthy that support appears. Recent work shows that explanations can influence fairness perceptions and, through these perceptions, shape users' tendency to follow AI recommendations, even when the explanations do not help them distinguish correct from incorrect outputs [3]. In parallel, research on fair multi-agent systems shows that efficiency-oriented optimisation may produce inequitable outcomes unless fairness is treated as an explicit design objective [4].

In HAI, these concerns are especially relevant when differences in transparency, system behaviour, or social cues affect how different users perceive and respond to the same agent. This is particularly important from a gender-fairness perspective, since prior work has shown that robot behaviour and identity cues can interact with gendered expectations and stereotypes in ways that shape user attitudes and trust [5, 6]. Under such conditions, some users may over-trust the system

because it appears reliable and well-aligned with their expectations, while others may under-trust it because they experience mismatch, discomfort, or exclusion. Such divergence weakens the legitimacy, acceptance, and overall effectiveness of the system as a whole.

Following these insights, D6.3 treats fairness as a socio-technical requirement rather than as a single metric. Unfairness and bias may emerge at multiple points in the system lifecycle, including data collection and representation, learning and decision policies, interaction design, and operational monitoring and governance [7, 5, 8, 4]. In the context of WP6, this is particularly relevant because gender fairness in HAI is shaped not only by algorithmic properties, but also by how uncertainty, explanations, control, and system limitations are communicated to users during interaction.

More specifically, gender-related effects may enter HAI through both system design and behaviour and user perception. Learned components may be trained on data that over-represent some groups or interaction styles, leading to transparency and explainability mechanisms that are more interpretable for some users than for others [9]. In addition, embodied or personified agents often carry gender-coded cues, such as voice, naming, persona, or embodiment, which can affect perceived competence, authority, warmth, and trust [10]. These issues are especially important in systems intended to support understanding, collaboration, and decision-making, because they shape not only user experience but also the reliability and legitimacy of the awareness support provided. More broadly, integrating sex and gender analysis into research and innovation has been shown to increase both research quality and societal relevance [11, 8]. This deliverable therefore treats gender fairness in two complementary ways, namely as a study-design and evaluation requirement and as a mechanism-design requirement.

Within SymAware, these questions are addressed through a user-centred, iterative methodology involving focus groups, Human-Robot Interaction (HRI) studies, and participatory design. D6.3 examines fairness through concrete mechanisms and empirical evidence, organised as follows:

- **Chapter 2** translates ethical guidance into operational fairness mechanisms (privacy, transparency, adaptability, and inclusion) relevant to mixed-autonomy interaction.
- **Chapter 3** presents survey and study results, reviewing how risk is identified in embodied AI and the extent to which current evidence supports fairness-sensitive evaluation.
- **Chapter 4** focuses on gender fairness practices within the project's own studies, proposing good practices for future HAI and MASs research.

Overall, this deliverable provides evidence that MASs cannot reliably support trustworthy situational awareness unless fairness is integrated into both the computational pipeline and the HAI mechanisms, validated through transparent and methodologically robust evaluation.

Design Guidelines for Gender Fairness

2.1 Introduction

Task 6.3 focuses on identifying and designing *mechanisms to promote fairness and prevent bias* in computational methods and interaction designs that support ethical and trustworthy awareness in HAI. This chapter addresses that objective by translating design guidance for trustworthy HRI into concrete fairness mechanisms that can be implemented, evaluated, and monitored in MASs.

We draw on a guideline set from the HRI literature that targets trust and acceptance through socio-technical design principles, including an Actor-Network Theory (ANT) perspective that treats robotic systems not only as tools, but also as *social mediators* shaped by and shaping their deployment context [12, 13, 14]. This perspective is directly relevant to WP6 because fairness in mixed-autonomy systems depends not only on technical performance, but also on how system behaviour is perceived, interpreted, and experienced by users. It also aligns with broader trustworthy-AI frameworks, which treat fairness, transparency, human oversight, and accountability as interdependent requirements rather than isolated properties [15, 16].

From a gender-fairness perspective, bias may arise through data collection practices, recognition modules, transparency failures, rigid autonomy settings, or interaction designs that assume a default user profile. For this reason, fairness must be considered across the full interaction pipeline, including data handling, learning, communication, adaptation, and run-time governance. This broader framing is also consistent with recent work in robotics that understands fairness as a multidimensional socio-technical concern shaped by legal, social, and technical factors [17].

The chapter interprets the proposed design guidelines as practical fairness mechanisms. Although they were originally formulated to support trustworthy and acceptable HRI more broadly, they also identify concrete points where gender-related bias, exclusion, or unequal support can be prevented or mitigated in MAS-supported interaction.

2.2 Mapping Design Guidelines to Gender Fairness

This section presents a structured set of design guidelines for robotic systems that aim to support effective performance, trust, security, and positive user experience. The guidelines are organised into five categories, inspired by the framework of [18] and the trustworthy and acceptable checklist of [19]. Together, they provide a practical basis for designing systems that respond to the needs of users in diverse contexts.

Applying the ANT perspective to HRI highlights that robotic systems participate in shaping

social relations. Assistive robots in healthcare, for example, do not only perform tasks, but may also influence trust and willingness to rely on the system through personalised interaction. Likewise, autonomous vehicles interacting with pedestrians use communicative interfaces to coordinate action. Conceptualising robots as social actors therefore helps identify where fairness-relevant effects may emerge in interaction.

The five categories discussed below are security and privacy, trust and transparency, safety and performance, competence and control, and positive experience. Taken together, they show that gender fairness in MAS-supported interaction depends not only on final outcomes, but also on how systems collect data, communicate decisions, adapt to users, protect safety, and support inclusion over time.

2.2.1 Security and Privacy

The deployment of robotic technologies raises concerns related to safety, data protection, and human rights. The ethical guidelines outlined by the European Commission[1] identify seven requirements for trustworthy artificial intelligence, including human agency, technical robustness, privacy, transparency, diversity, societal well-being, and accountability. For robotic systems that process personal data, the EU General Data Protection Regulation (GDPR)[2] requires user consent for data processing and guarantees the right to withdraw that consent at any time.

From a gender-fairness perspective, privacy and security are also safeguards against profiling and differential exposure. Unnecessary collection or inference of sensitive information may increase the risk that users are categorised or treated differently on the basis of demographic attributes.

- **DG1** — *Comply with data protection regulations and promote transparent documentation.* Designers and developers should comply with data protection regulations, such as the GDPR in the EU, in order to enhance confidentiality. They should clearly communicate what information is being collected and how it will be used by the robotic system, supported by transparent documentation and strict hardware and software security policies. Integrity should be prioritised through authentication measures and encryption protocols that prevent unauthorised access and data breaches.

- **DG2** — *Implement data minimisation and anonymisation.* Developers should implement measures to ensure that collected data remain anonymous and cannot be linked to specific individuals. Robotic systems should prioritise only the data necessary for task completion, thereby limiting the collection of personal data to what is strictly required for functionality and minimising intrusion into users' privacy. Personal identification should occur only with explicit consent. Where distinction between users is necessary, the system should rely on pseudonymous identifiers whenever possible.

DG1 and DG2 are fairness-relevant because they reduce unnecessary demographic exposure and make data practices more auditable. Fairness, however, also depends on whether users can understand how the system operates, which brings the discussion to trust and transparency.

[1] https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai
[2] https://gdpr-info.eu

2.2.2 Trust and Transparency

Trust and transparency are closely linked to the principles of security and privacy. Beyond protecting data, a trustworthy system should make its behaviour, capabilities, and limitations understandable to users.

This is important for gender fairness because unequal treatment may remain hidden when users cannot interpret how the system perceives, classifies, or responds to them. Transparent interaction supports scrutiny and makes it easier to identify whether support or recognition quality differs across user groups.

- **DG3** — *Foster transparent communication.* Designers and developers should prioritise transparent communication regarding the robotic system's path, movement, states, plans, and autonomy, and should implement features that enhance user comprehension and control throughout interaction. This includes making object and person recognition transparent and comprehensible through mechanisms that allow users to understand how the robot identifies and interacts with objects and individuals in its environment.

- **DG4** — *Design for adaptability, feedback processing, and environmental responsiveness.* Trust is fostered when the robotic system adapts its interaction to different environments through multimodal approaches and intuitive communication mechanisms. The system should provide feedback on user operations, reassure users before executing actions, and communicate limitations, errors, and reliability dynamics in a timely manner. This adaptability should extend to processing environmental feedback and responding to user preferences and needs, thereby ensuring dynamic and context-aware interaction.

- **DG5** — *Incorporate the appropriate level of human likeness or anthropomorphism.* Developers should carefully integrate human-like features only when they serve a clear purpose and align with the intended context, thereby supporting an appropriate degree of trust. The robotic system's appearance and behaviour should remain consistent with its actual functions in order to avoid creating false expectations.

DG3–DG5 support gender fairness by improving intelligibility, accommodating diverse interaction needs, and reducing the risk that anthropomorphic cues reinforce stereotypes or misleading expectations. Yet fairness also depends on how systems protect users in shared environments.

2.2.3 Safety and Performance

Safety is a core requirement for trustworthy robotic systems operating in shared environments. Such systems must avoid harm while maintaining reliable performance under diverse conditions.

From a gender-fairness perspective, safety and performance should not be calibrated around a narrow or assumed default user. A system may perform well on average while still producing unequal burdens if detection, spacing, timing, or responsiveness work less effectively for some users than for others.

- **DG6** — *Ensure physical and emotional safety of users.* Developers should devise and implement dedicated measures to ensure users' physical and emotional safety while also enhancing

the performance of the robotic system. This may include incorporating advanced sensor technology for real-time monitoring and detection of relevant agents, both human and non-human, in the immediate environment. Safety algorithms should include features such as automatic emergency braking and perceptible, proactive collision avoidance.

- **DG7** — *Incorporate social adaptability during the interaction.* The robotic system should dynamically adjust its behaviour to prioritise user safety when users enter its operational space. It should uphold social norms and demonstrate cultural sensitivity through friendly and polite communication that facilitates task completion. Physical contact should be permitted only when it is relevant to the task and explicitly allowed by the user, thereby coordinating assertive behaviour with user preferences. **DG7** is closely connected to **DG4**, as both emphasise adaptability to different situations, user needs, and environments, which contributes significantly to user safety.

DG6 and DG7 are fairness-relevant because they extend safety beyond technical avoidance of harm to include emotional safety, social comfort, and respect for user boundaries. In addition to being safe, a fair mixed-autonomy system should also support meaningful user agency.

2.2.4 Competence and Control

The acceptance of robotic systems is closely tied to their ability to perform tasks accurately, support user efficacy, and enable seamless collaboration in socio-technical environments. Different levels of autonomy are therefore crucial for effective and adaptable system behaviour. Reinmund and colleagues [20] suggest that variable autonomy can be understood along four dimensions—initiative, specificity, flexibility, and context—thereby providing useful guidance for robotics.

From a gender-fairness perspective, competence and control matter because unequal agency may arise when autonomy settings are rigid, when user preferences are not sufficiently recognised, or when some interaction styles are implicitly privileged over others.

- **DG8** — *Design for task management and task integration in sociotechnical systems.* In collaborative task scenarios involving multiple actors, designers should implement user-configurable task coordination. This enables users to customise and coordinate tasks according to their specific needs and contextual requirements, for example by prioritising critical tasks before execution and establishing clear communication channels through which anomalies or uncertainties can be addressed. In line with the ANT perspective, this guideline promotes seamless task integration within socio-technical systems, supporting adaptability and user-driven customisation for improved collaboration.

- **DG9** — *Incorporate context-aware autonomy levels and flexibility.* Designers should carefully select the appropriate level of autonomy in light of the context in which the robot will operate. This decision affects the overall performance of the human–robot team, user experience, and interaction with others. Choosing between human-, system-, and mixed-initiative implementations involves balancing factors such as human control, efficiency, and consistency. It is therefore essential to align autonomy levels with deployment requirements while allowing flexibility in autonomy changes to meet user needs and improve transparency in system operation.

DG8 and DG9 support gender fairness by preserving user agency and reducing the risk that control structures systematically benefit some users more than others. Still, fairness must also be reflected in whether interaction is respectful, inclusive, and sustainable over time.

2.2.5 Positive Experience

A positive user experience is an important condition for trust, acceptance, and sustained engagement with robotic systems. In socio-technical environments, this depends not only on usability, but also on whether users feel respected, included, and appropriately supported.

This matters for gender fairness because a system may perform correctly while still producing stereotypical, exclusionary, or inaccessible interactions. Fairness therefore also concerns dignity, accessibility, and equal respect in the design of the overall user experience.

- **DG10** — *Ensure respect for human rights, diversity, and inclusion.* Designers, developers, and organisations must ensure that robotic systems respect human rights and dignity by adhering to ethical principles and avoiding behaviours that compromise individuals' fundamental rights. When implementing AI algorithms, such as computer vision modules, accountability is essential. Incorporating and auditing algorithms that actively prevent bias and stereotypes helps promote fairness across interactions. Robotic systems should also be capable of interacting with users from different backgrounds and with different needs, for example by adapting behaviour to vulnerable or impaired individuals.

- **DG11** — *Design for long-term use and novelty effect mitigation.* Designers should aim for long-term usability and acceptance in order to overcome novelty effects and sustain user engagement over time.

DG10 makes the fairness dimension explicit through diversity, inclusion, and bias prevention, while DG11 highlights that fairness should also be examined over time, since longer-term use may reveal disparities that are not visible in short-term evaluations.

2.3 Bridge to Chapters 3 and 4

The guidelines discussed above serve not only as design recommendations, but also as fairness-relevant mechanisms. They identify where gender-related disparities may emerge and how such risks can be mitigated in practice, whether through privacy-preserving data practices, transparent communication, adaptive interaction, inclusive design, or long-term evaluation.

The mapping presented above provides the mechanism catalogue required by Task 6.3. Chapter 3 evaluates which mechanisms are most relevant based on project evidence, including risk measures, user samples, findings, and fairness risks in datasets used for training or demonstration. Chapter 4 consolidates gender fairness practices across project user studies and specifies how gender considerations are embedded in mechanism selection and evaluation protocols.

2.4 Conclusion

This chapter has shown that the design guidelines developed for trustworthy HRI can also be used as concrete fairness mechanisms for mixed-autonomy systems. Taken together, they provide a

practical basis for addressing gender fairness across multiple dimensions of system design, including privacy-preserving data handling, transparent communication, adaptive behaviour, user control, inclusive interaction, and long-term use.

A central insight is that gender fairness in MAS-supported interaction is cumulative. It depends on the combined effect of many design decisions rather than on any single intervention. Data minimisation and documentation help reduce unjustified exposure. Transparency and adaptability make system behaviour more understandable and contestable. Safety and autonomy design influence how protection and control are distributed. Diversity- and inclusion-oriented design helps ensure equal respect across users.

More broadly, the chapter highlights that gender fairness in HAI should not be understood narrowly as a binary or purely statistical property. It also concerns whether systems are designed and evaluated in ways that acknowledge diverse experiences, avoid reinforcing stereotypes, and remain sensitive to different needs and vulnerabilities. The next chapter builds on this foundation by examining empirical evidence from surveys, user studies, and datasets in order to assess how fairness-related concerns appear in practice.

Systematic Review on Risk Assessment Relevant to Fairness

3.1 Introduction

This chapter presents the empirical evidence relevant to fairness in SymAware, with particular attention to how risk is identified, measured, and interpreted in embodied HRI. In line with Task 6.3, it examines both the methodological landscape of risk in embodied AI and the extent to which current evidence supports fairness assessment, including gender fairness. The chapter is motivated by the observation that fairness in HAI depends not only on mechanism design, but also on the quality of the empirical evidence used to evaluate harms, representativeness, and differential impact across user groups.

The chapter is organised in two parts. First, it reviews studies on risk in embodied AI and analyses their methods, metrics, harms, drivers, and alignment with the EU Trustworthy AI requirements [15]. Second, it examines what this evidence reveals about fairness-sensitive evaluation, especially in relation to participant diversity, bias, discrimination, and dataset limitations. Together, these analyses identify the main strengths and weaknesses of the current evidence base and provide the empirical grounding for the recommendations developed later in Chapter 4.

3.2 Methods

3.2.1 Review scope and research questions

This review examines how risk is identified, measured, and minimised in HRI involving embodied AI. Its scope was defined using the Population, Concept, and Context framework [21, 22]. The population comprised humans who were co-present with, or directly affected by, embodied AI systems, including industrial workers, patients and care recipients, and users in domestic or public environments. The concept of interest was risk in embodied HRI, including physical hazards and socio-cognitive risks such as trust miscalibration, bias, cognitive load, and loss of agency. The context covered both laboratory-based experiments and more deployment-relevant evaluations in real or semi-real environments.

Embodied AI systems in HRI can generate risks that extend beyond task failure, including physical strain, psychological burden, unfair social behaviour, and wider societal harms. Existing studies report a wide range of performance, usability, and model-based metrics, but these are rarely

synthesised through a shared risk vocabulary. To address this gap, the review organises evidence around a unified extraction schema linking *methods*, *risk metrics*, *failure modes*, *harms*, *risk drivers*, and *Trustworthy AI evidence*.

The review was guided by the following research questions

- **RQ1.1** What methodologies are used to study risk in HRI involving embodied AI
- **RQ1.2** What metrics and evaluation approaches are used to identify possible failures and support risk minimisation in embodied AI HRI systems
- **RQ1.3** How do these risks relate to harms
- **RQ2** Which robot, human, and contextual factors drive risk in embodied AI
- **RQ3** To what extent do existing embodied AI implementations address ethical principles and trustworthiness requirements as defined in the EU Trustworthy AI Guidelines

3.2.2 Eligibility criteria

Studies were eligible for inclusion if they met four criteria. First, they had to examine *physically embodied AI systems*, such as robots, that interact with their environment through sensing and actuation, thereby enabling perception, action, and adaptation. Second, they had to involve *HRI*, including task collaboration, communication, or shared environments in which robot behaviour could affect human experience or outcomes. Third, they had to address at least one dimension relevant to *Trustworthy AI*, such as robustness, safety, accountability, fairness, transparency, privacy, or human oversight [15]. Fourth, they had to provide *risk-relevant evidence*, either *ex-ante*, for example through design, testing, or mitigation frameworks, or *ex-post*, for example through incidents, failures, misuse, or operational harms arising during deployment.

Studies were excluded if they did not involve real HRI, for example perception-only evaluations, video-only studies, simulation, or purely technical benchmarks without a human interaction component. Pure teleoperation and Wizard-of-Oz studies were also excluded unless they explicitly reported risk, failure, or mitigation evidence relevant to embodied AI deployment.

3.2.3 Information sources, search strategy, and study selection

The search strategy combined three concept blocks. The first targeted risk in AI systems using terms such as *risk*, *hazard*, *incident*, *failure*, *near miss*, *safety*, *harm*, and *liability*, combined with *AI* or *artificial intelligence*. The second targeted embodied HRI using terms such as *robot*, *human–robot interaction*, *HRI*, *social robot*, *cobot*, *assistive robot*, and *service robot*. The third targeted ethics- and governance-related dimensions of Trustworthy AI, including robustness, safety, privacy, data governance, transparency, explainability, fairness, bias, discrimination, inclusiveness, accountability, and human oversight.

Searches were conducted in Scopus, Web of Science, and the ACM Digital Library. Forward and backward citation searching, together with hand-searching of relevant HRI and robotics venues, was also undertaken. The final search was conducted on 10 September 2025. The database searches yielded 8,196 records in total, comprising 4,812 from Scopus, 3,140 from Web of Science, and 244 from the ACM Digital Library. After deduplication, 6,047 unique records remained. Screening was

supported by ASReview as a human-in-the-loop active-learning tool to prioritise title and abstract screening. After 306 prioritised records had been screened and the stopping rule applied, 74 records were assessed at full text, of which 34 met the inclusion criteria.

3.2.4 Data extraction framework

To answer the research questions, we developed a coding scheme for data extraction and analysis. The scheme was refined iteratively among the co-authors to clarify category boundaries, resolve ambiguities, and improve consistency in the interpretation of risk constructs. For RQ1.1, RQ1.3, RQ2, and RQ3, data were extracted at the study level. For RQ1.2, data were extracted at the metric level because individual studies often reported multiple measures for the same risk construct and these measures could serve different analytic functions.

This distinction allowed us to separate outcomes treated as evidence of risk from the signals used to detect, predict, or quantify those outcomes. Harm coding mapped study-specific risk framings onto the human-centred taxonomy of harms proposed by [23]. Driver coding captured robot-, human-, context-, and AI-related factors so that risk could be analysed as a socio-technical phenomenon rather than as a property of the robot alone. Trustworthy AI alignment was coded using a conservative evidence threshold so that governance claims were recorded only where supported by explicit or clearly inferable design and evaluation details.

Missing information was coded as NR rather than inferred. Extraction was conducted by one reviewer and checked through discussion of inconsistencies with the wider review team.

3.3 Results

3.3.1 Identification and Evaluation of Risk

Methodological landscape for risk in embodied AI RQ1.1

To address RQ1.1, we characterised the methodological landscape of the included embodied HRI risk literature ($n = 34$), focusing on application domains, study designs, ecological realism, demographic reporting, and evaluation approaches.

Risk-focused embodied HRI research was concentrated in industrial contexts (50.0%, $n = 17$), with substantially smaller representation in social HRI and healthcare, each at 14.7% ($n = 5$). Mobility and domestic deployments each accounted for 5.9% ($n = 2$). Public spaces and educational settings were rare, and the remaining domains were grouped as “other” categories (8.8%, $n = 3$). Methodologically, the field relied primarily on controlled experimental designs (85.3%, $n = 29$), with mixed-method studies (8.8%, $n = 3$) and purely technical evaluations (5.9%, $n = 2$) appearing far less often. Overall, risk in embodied HRI is most often studied in controlled interactions and in relatively structured domains.

Our findings also reveal a realism–readiness gap. Across the corpus, ecological realism was predominantly low, with 79.4% of studies coded as low realism ($n = 27$), compared with 14.7% at moderate realism ($n = 5$) and 5.9% at high realism ($n = 2$). This indicates that risk is usually examined under tightly controlled conditions rather than in operational contexts that expose systems to interactional and environmental variability. The evidence base is therefore weighted towards laboratory measurement, with limited coverage of deployment-relevant dynamics.

Table 3.1: Cross-tabulation of evaluation approach categories, coded instances $n = 45$, by ecological realism

Evaluation approach category	**Ecological realism**			**All**
	Low	**Moderate**	**High**	
Analytic Risk	27	5	1	33
Field Safe	0	1	2	3
Formal Verification	1	0	0	1
Red Teaming	2	0	0	2
Safety Assurance	0	1	0	1
Stress	4	1	0	5
All coded instances	34	8	3	45

Note — categories are non-exclusive and counts represent coded instances rather than unique papers

Concerning demographic reporting, the total number of participants across all included studies was 733. However, three studies did not report the total number of participants. Restricting the analysis to the 31 studies that reported sample size, demographic reporting remained incomplete. Eleven studies (32.4%) did not report gender distribution and 15 (44.1%) did not report the age range of participants. Among studies reporting gender, aggregated participation was male-skewed, with 57.4% men ($n = 381$) and 42.6% women ($n = 283$), and no non-binary participants were reported (total reported $n = 664$). Overall, reporting of participant characteristics critical for assessing generalisability and fairness-related heterogeneity was inconsistent across the corpus.

From a gender-fairness perspective, these reporting gaps are important because they make it difficult to determine whether observed risk patterns generalise across user groups or whether some risks may be disproportionately borne by underrepresented participants. The absence of reported non-binary participants further indicates that current embodied HRI risk research continues to rely on limited demographic categories.

To provide a comparable high-level characterisation of evaluation methods, we coded each study using one or more evaluation approach categories stored as `method_tags` in the extraction schema. These categories were non-exclusive, which yielded 45 coded instances across the identified studies.

The distribution of evaluation approach categories was highly skewed. *Analytic Risk* accounted for 33 out of 45 coded instances, whereas deployment-oriented monitoring aligned with *Field Safe* was uncommon, with 3 out of 45. *Stress* contributed 5 instances and *Red Teaming* 2. Formal assurance-oriented approaches were rare, with *Safety Assurance* and *Formal Verification* each appearing once. Stratifying by ecological realism showed that *Analytic Risk* was concentrated under low realism, whereas *Field Safe* appeared only in more operationally oriented studies, see Table 3.1. Overall, the evidence base emphasises interaction outcomes under laboratory-style evaluations, with limited use of deployment-linked approaches capable of capturing operational risk signals under realistic conditions.

Metrics identify failures and support risk minimisation RQ1.2

To address RQ1.2, we applied a two-stage coding procedure to harmonise risk operationalisations across studies. We first extracted verbatim definitions of risk and corresponding justifications, then derived a shared set of ten risk themes through interpretive analysis. We then constructed a

rule-based codebook using predefined lexical indicators and word stems specified *a priori*. Excerpts matching more than one theme were manually reviewed in order to assign a primary theme for metric-level aggregation.

A total of 251 risk metric instances were extracted from the included studies. Each instance captured the functional role of the metric, the timing of risk assessment, the indicator type, the measurement modality, the detection route, and whether the evidence primarily related to technical system behaviour or human-facing outcomes. Each metric instance was linked to one of the ten risk themes.

Risk themes, functional role, and the assessment pipeline Figure 3.1 maps risk themes to metric role, measurement modality, and metric type. Across all instances, physical and psychological safety was the most frequently represented theme (32.67%, $n = 82$), followed by cognitive load and performance (21.51%, $n = 54$). Trust calibration and over- or under-reliance accounted for 12.75% ($n = 32$), system robustness and perception failure for 11.55% ($n = 29$), and transparency, mental models, and explanation for 9.16% ($n = 23$). Less frequent themes included bias, discrimination, and stereotyping (4.78%, $n = 12$), public health and hygiene compliance (2.39%, $n = 6$), psychosocial wellbeing and user discomfort (2.39%, $n = 6$), cooperative fairness (1.99%, $n = 5$), and loss of human agency and autonomy (0.80%, $n = 2$).

From a fairness perspective, this distribution shows that themes directly related to bias, discrimination, stereotyping, cooperative fairness, and loss of agency were comparatively rare. Fairness-related risks may therefore be under-detected not because they are absent, but because they are less often operationalised as primary objects of measurement.

Outcome metrics dominated the evidence base, making up 63.75% of instances ($n = 160$). Detection input metrics contributed 18.33% ($n = 46$), while model performance metrics constituted 17.93% ($n = 45$). Objective measurement predominated at 77.69% of instances ($n = 195$), whereas subjective measurement comprised 22.31% ($n = 56$). Among metric types, model-based metrics were the single largest category (25.90%, $n = 65$), closely followed by behavioural or performance measures (25.10%, $n = 63$). Subjective and qualitative approaches were comparatively uncommon.

Timing and indicators Figure 3.2 maps risk themes to metric role, risk timing, indicator type, and risk focus. Ex-post metrics were most prevalent, comprising 65.74% of instances ($n = 165$), followed by ex-ante metrics at 24.70% ($n = 62$), while metrics spanning both phases accounted for 9.56% ($n = 24$). Leading indicators dominated at 74.10% ($n = 186$), whereas lagging indicators comprised 25.90% ($n = 65$). The risk-focus layer was more evenly distributed, with technically focused instances slightly exceeding human-focused ones.

Figure 3.3 summarises metric-instance counts by risk theme and joint role–timing category. Several human-facing themes were assessed predominantly *ex-post*, including trust calibration and over- or under-reliance, transparency, mental models, and explanation, and bias, discrimination, and stereotyping. By contrast, system robustness and perception failure showed a more proactive profile. Physical and psychological safety occupied an intermediate position. Taken together, these findings suggest that assessment remains largely *ex-post*, especially for human-centred themes, whereas more proactive assessment is concentrated in technical failure pathways.

Finally, we extracted the detection route for each metric. Log-based routes were most prevalent, representing 37.45% of instances ($n = 94$). Survey- or instrument-based routes accounted for

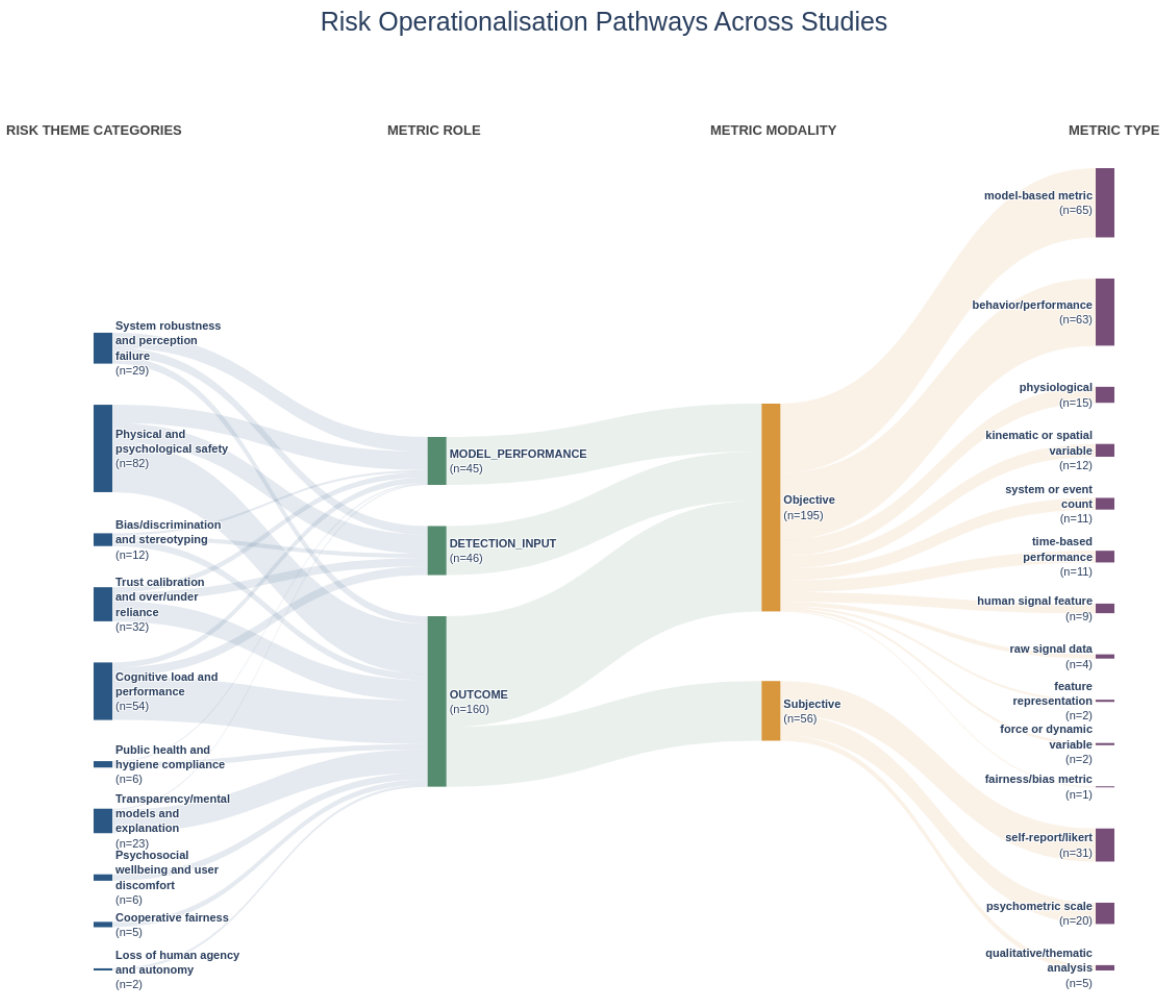

Figure 3.1: Methodological operationalisation pathways of HRI risk. Sankey diagram mapping risk themes to metric role, measurement modality, and metric type. Link widths are proportional to the number of risk metric instances, $N = 251$, jointly coded with the connected categories

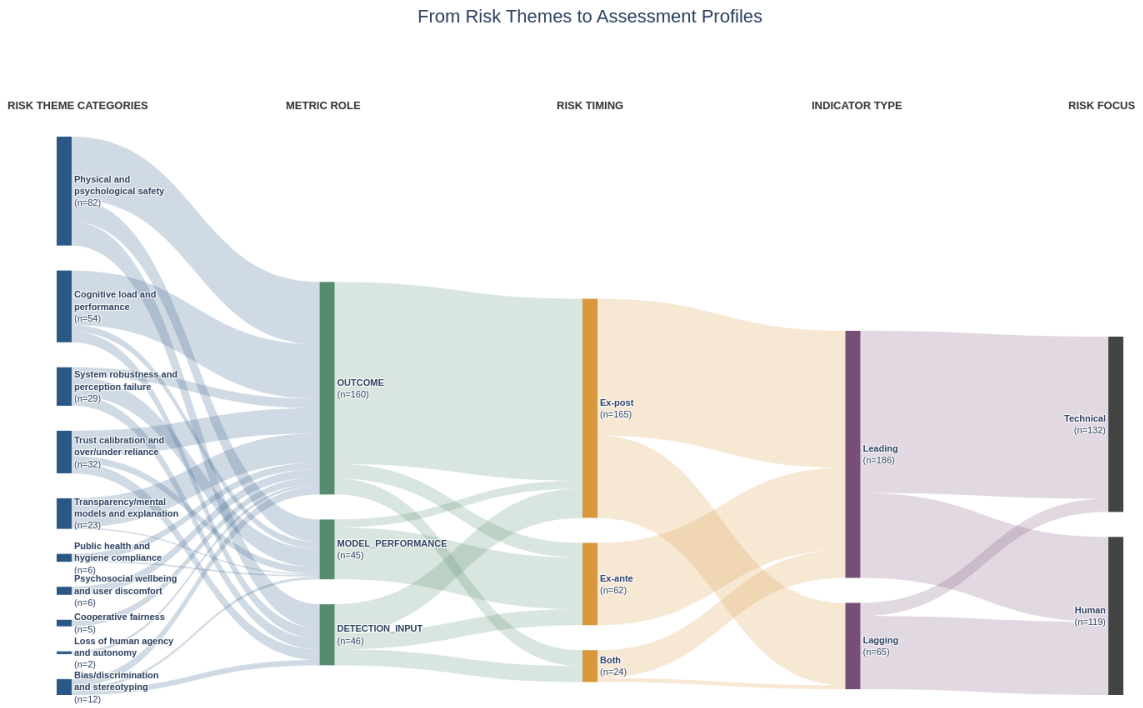

Figure 3.2: Proactivity and focus in HRI risk assessment. Sankey diagram mapping risk themes to metric role, risk timing, indicator type, and risk focus. Link widths are proportional to the number of risk metric instances, $N = 251$, jointly coded with the connected categories

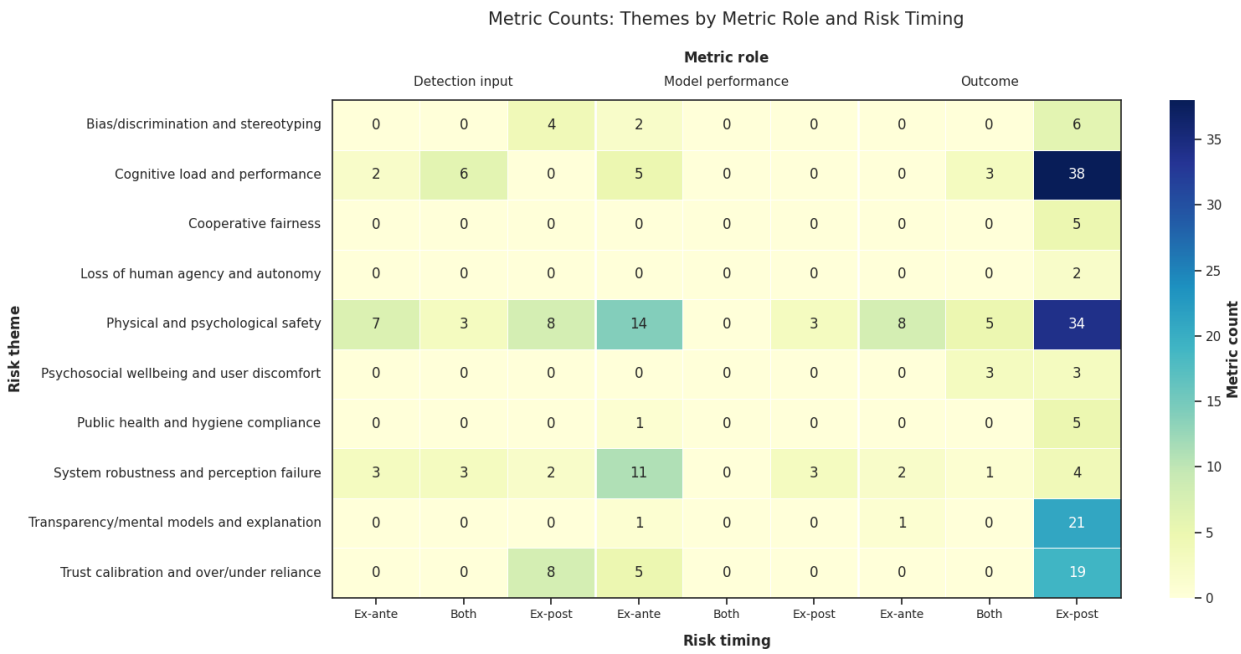

Figure 3.3: Distribution of extracted risk metric instances across themes and operationalisation profiles. Cells report counts for each risk theme by metric role and risk timing, highlighting concentrations of evidence and sparsely represented combinations

23.11% ($n = 58$), sensor-based routes for 20.32% ($n = 51$), and mixed routes for 13.55% ($n = 34$). Observer-based routes were least common, representing 5.58% of instances ($n = 14$).

Harms and their relation to risk operationalisation RQ1.3

This research question examined the association between the technical risk themes identified in our corpus and real-world harms using the human-centred taxonomy proposed by [23]. Each study was mapped to the framework at two levels, namely *Harm Taxonomy*, representing high-level categories, and *Harm to Subject*, representing specific consequences.

Multi-harm mappings were common, with 70.59% of studies ($n = 24$) linked to two or more harms. This process yielded a *Harm Registry* of 64 distinct entries. *Physical Harm* ($n = 28$) and *Psychological Harm* ($n = 22$) were the primary consequences represented in the corpus, together accounting for more than 78% of all entries. *Societal and Cultural Harm* ($n = 9$) formed a secondary layer of impact, while other categories were much rarer.

At the *Harm to Subject* level, bodily injury ($n = 15$) and personal health deterioration ($n = 10$) were the most common consequences within *Physical Harm*. *Psychological Harm* was the most diverse category, with anxiety or distress ($n = 9$), emotional distress ($n = 7$), over-reliance ($n = 4$), and coercion or manipulation ($n = 2$) all appearing. Within *Societal and Cultural Harm*, information degradation was the leading subject ($n = 5$), followed by stereotyping ($n = 2$) and damage to public health ($n = 2$). *Human Rights and Civil Liberties* harms were infrequent and related mainly to discrimination.

Although physical and psychological harms dominated the corpus, the relative scarcity of harms

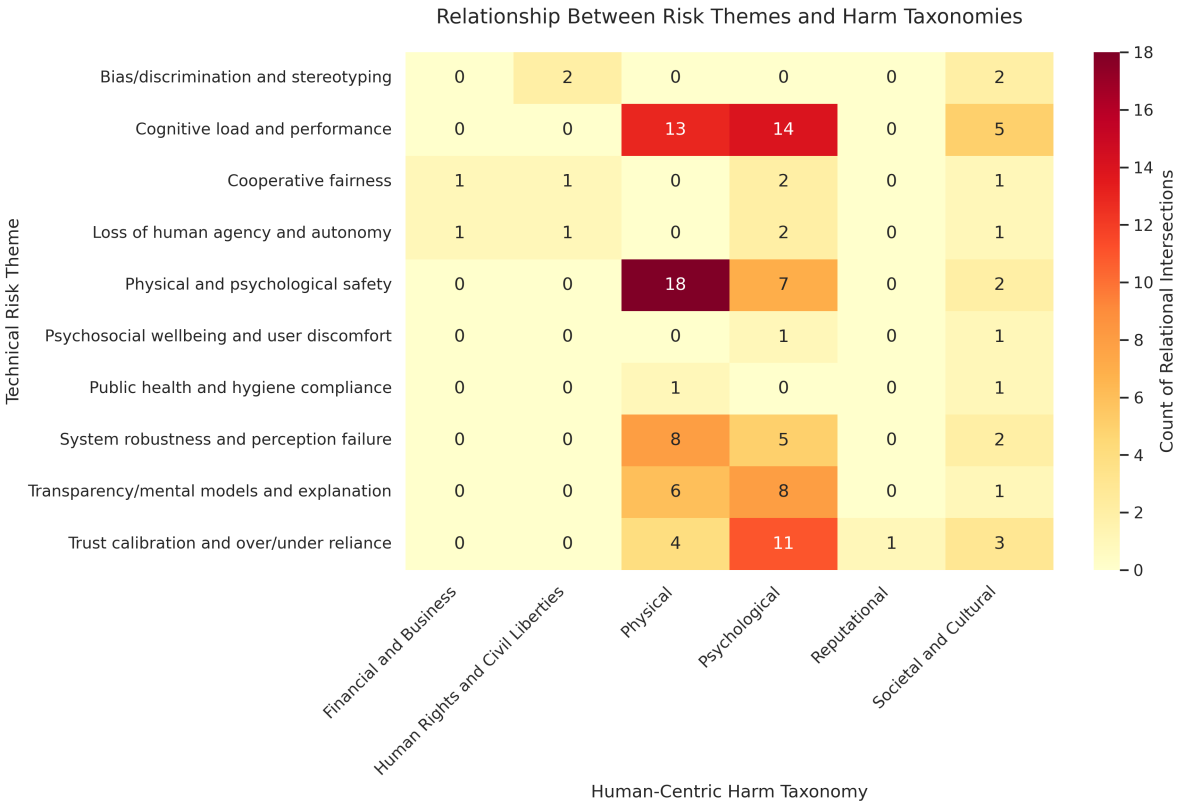

Figure 3.4: Heatmap showing the frequency of unique relationships between risk themes and high-level human-centred harm taxonomies across the included studies. Darker cells indicate more frequently observed risk–harm combinations. Physical and psychological harms were the most frequently represented taxonomies

linked to discrimination, stereotyping, or civil liberties should be interpreted cautiously. These harms are often more difficult to operationalise within standard HRI evaluation protocols and may therefore remain underreported. From a gender-fairness perspective, this suggests that current harm assessments capture immediate safety consequences more reliably than unequal or exclusionary effects emerging through social interaction or repeated exposure over time.

To identify risk–harm trajectories, the extracted harms were merged at the study level with the standardised risk themes described in Section 3.3.1. The resulting dataset contained 126 unique relational intersections. Figure 3.4 illustrates the frequency of observed relationships between risk themes and high-level harm taxonomies. Overall, the results suggest that technical vulnerabilities do not produce isolated failures, but may instead cascade across multiple harm categories.

The strongest alignment was found between *Physical and Psychological Safety* risks and *Physical Harm* ($n = 18$ overlaps). A second prominent pattern was the overlap between *Cognitive Load and Performance* and both *Psychological Harm* ($n = 14$) and *Physical Harm* ($n = 13$). Finally, technical risks related to *Trust Calibration* mapped strongly onto *Psychological Harm* ($n = 11$), largely through mechanisms of user over-reliance on faulty system outputs.

3.3.2 Drivers of Risk in Embodied AI RQ2

For RQ2, we synthesised the risk drivers identified across the included studies at the robot, human, and contextual levels, and characterised the AI-based technical components embedded in the reviewed systems.

Robot-, human-, and context-related factors

Using the framework established in [24], we examined risk in embodied AI as a socio-technical phenomenon. Robot-related drivers included both attribute-related and performance-related factors. Attribute-related factors included design features such as anthropomorphism and morphology, identified as key drivers in 17.6% of studies. We also considered physical and social characteristics, including robot type and embodiment. Industrial or collaborative robotic arms were the most frequently represented platform (61.8%, $n = 21$), followed by social or humanoid robots (20.6%, $n = 7$). Performance-related factors included predictability, transparency, and reliability, identified as the main drivers in 23.5% of studies. Across studies, transparency mechanisms such as contrastive verbal explanations and SHAP-based visualisations emerged as important means of calibrating trust and reducing over-reliance in safety-critical decision-making [25, 26, 27, 28].

Human-related drivers were divided into characteristic-based and ability-based factors. Characteristic factors were the category most frequently reported, accounting for 32.4% of the studies ($n = 11$). These centred primarily on gender and age. In social HRI, gender shaped susceptibility to biased robot behaviour that amplified stereotypes [6, 5], whereas age and digital literacy influenced the psychosocial burden of companion robots in domestic settings [29]. Other traits, such as dispositional trust and anxiety, also shaped risk. Ability-based factors accounted for 11.8% of studies ($n = 4$), with cognitive workload and attentional capacity emerging as key constraints on safe collaboration [30, 31].

These findings are especially relevant for gender fairness because they show that user characteristics are not merely background variables, but can shape how risks emerge and how systems are experienced. Where gender influences exposure to biased robot behaviour, stereotype amplification, or differential interaction patterns, it should be treated as a substantive factor in risk analysis.

Context-related drivers defined the spatial and operational boundaries of the interaction. Team and collaboration factors included proximity, co-location, and shared understanding of workspace boundaries. Task-related factors included task complexity and features of the physical environment, such as hazardous materials or ambient household noise, which increased failure severity [32, 29]. These findings underscore the need for adaptive risk management that accounts for the physical environment as a determinant of system reliability.

AI mechanisms and data foundations

We summarised the extracted technical descriptors reported at the study level ($N = 34$) across four dimensions, namely system layer, AI model category, input data types, and dataset issues, see Table 3.2. These dimensions were coded as non-exclusive when studies reported multiple layers or modalities.

At the system layer, risk-related AI components were concentrated primarily in the perception layer (32.35%, $n = 11$) and the control layer (26.47%, $n = 9$). The planning and dialogue layers

were each represented in 20.59% of studies ($n = 7$), whereas recommendation mechanisms were rare (5.88%, $n = 2$).

Regarding algorithmic architecture, rule-based or heuristic approaches were the most frequently reported (32.35%, $n = 11$), followed by deep learning methods (26.47%, $n = 9$) and classical machine-learning approaches (23.53%, $n = 8$). In 17.65% of studies ($n = 6$), the model type was not explicitly reported.

Input data types were reported inconsistently, and 38.24% of studies did not specify the inputs used ($n = 13$). Among those that did, vision or image data appeared in 20.59% of studies ($n = 7$), kinematic, spatial, or haptic signals in 17.65% ($n = 6$), and physiological or biosignal inputs in 17.65% ($n = 6$). Other input types, including text, metadata, and task-state representations, were reported in 23.53% of studies ($n = 8$).

Reporting of dataset issues was absent in 55.88% of studies ($n = 19$). Among the studies that identified such issues, lab-only data limitations were the most frequently coded primary issue (26.47%, $n = 9$), followed by data imbalance (8.82%, $n = 3$) and out-of-distribution or synthetic-data issues (8.82%, $n = 3$).

Table 3.2: AI mechanisms and data foundations reported across studies, $N = 34$. System layer and input data types are non-exclusive. AI model category and dataset issues are coded as a single primary label per study

Dimension	**Category**	**Studies (n)**	**Share (%)**
System layer	Perception	11	32.35
	Control	9	26.47
	Planning	7	20.59
	Dialogue	7	20.59
	Recommendation	2	5.88
AI model category	Rule-based or heuristic	11	32.35
	Deep learning or neural networks	9	26.47
	Classical ML or other	8	23.53
	Not reported	6	17.65
Input data types	Vision or image	7	20.59
	Kinematic, spatial, or haptic	6	17.65
	Physiological or biosignal	6	17.65
	Other or ambient	8	23.53
	Not reported	13	38.24
Dataset issues	Not reported	19	55.88
	Lab-only data	9	26.47
	Imbalance	3	8.82
	Out-of-distribution or synthetic	3	8.82

Overall, the extracted evidence indicates that drivers of risk in embodied AI are multi-factor, spanning robot embodiment and reliability, human workload and experience, and interaction context.

3.3.3 Trustworthy AI and governance alignment RQ3

To address RQ3, we examined the extent to which the included embodied AI implementations aligned with the EU Trustworthy AI Guidelines by coding each study against the seven Trustworthy AI requirements [15]. Coverage was recorded when a study reported a concrete design choice, evaluation component, or governance practice consistent with a given requirement.

Alignment was partial overall, see Table 3.3. Studies addressed a mean of 3.56 requirements (Mdn = 3.5, range = 1–6), and none covered all seven. Most studies addressed three or four requirements ($n = 23$), whereas only six addressed five or more.

Table 3.3: Coverage of EU Trustworthy AI requirements across the $N = 34$ studies, binary presence or absence

Trustworthy AI requirement	**Studies (n)**	**Percentage (%)**
Human agency and oversight	14	41.18
Technical robustness and safety	25	73.53
Privacy and data governance	21	61.76
Transparency	19	55.88
Diversity, non-discrimination, and fairness	3	8.82
Societal and environmental wellbeing	26	76.47
Accountability	13	38.24

Alignment was strongest for requirements most directly linked to safety and system performance. *Societal and environmental wellbeing* was the most frequently addressed requirement (76.47%, $n = 26$), followed by *Technical robustness and safety* (73.53%, $n = 25$). *Privacy and data governance* appeared in 61.76% of studies ($n = 21$), although this was usually addressed through procedural measures such as ethics approval and informed consent rather than through technical safeguards.

By contrast, governance-oriented requirements were addressed less consistently. *Accountability* was identified in 38.24% of studies ($n = 13$) and was usually operationalised through traceability practices such as logging and monitoring rather than stronger forms of auditability or responsibility allocation. *Diversity, non-discrimination, and fairness* remained rare (8.82%, $n = 3$), appearing only in studies that explicitly examined bias, stereotyping, or differential treatment [6, 5, 29].

Overall, current embodied AI implementations show three broad patterns, namely strong alignment with safety and wellbeing, moderate alignment with transparency and human agency, and limited coverage of fairness and accountability. This is especially significant for this deliverable because fairness concerns, including gender-related bias and unequal treatment, are still rarely operationalised as central evaluation targets in embodied HRI.

3.4 Summary of the Findings

The findings of this chapter show that current embodied AI and HRI research provides an uneven evidence base for trustworthy and fair human-agent interaction. The ALTAI-based synthesis indicates that governance-relevant evidence is already emerging, but that it remains distributed unevenly across the requirements for Trustworthy AI. Coverage was strongest for societal and environmental wellbeing and for technical robustness and safety, moderate for privacy and data

governance and for transparency, and notably weaker for human agency and oversight, accountability, and especially diversity, non-discrimination, and fairness. No study addressed all seven requirements. This suggests that aspects of Trustworthy AI aligned with established HRI concerns, such as safety, workload, and explainability, are easier to operationalise within current experimental paradigms than those requiring explicit governance mechanisms or attention to differential impact.

At the same time, the review identified some important signs of progress. In particular, several studies treated communication and explanation as functional risk controls supported by concrete metrics rather than as abstract ethical aspirations. This indicates that at least some dimensions of Trustworthy AI, especially transparency, are increasingly being operationalised in ways that are directly relevant to interaction quality and risk reduction [27, 28, 33]. However, more persistent limitations remain in relation to privacy, data governance, accountability, and fairness. Although privacy and data governance were coded relatively often, the supporting evidence was frequently procedural rather than technical. Accountability was also reflected more often in logging and documentation than in stronger forms of auditability or responsibility allocation. Fairness remained particularly underdeveloped. Together with limited reporting of dataset issues, this suggests that governance-relevant evidence is weakest where trustworthiness depends on lifecycle oversight and differential impact assessment [16, 15].

A second major finding concerns the limitations of the current evidence base with respect to fairness, representativeness, and identity-related risk. A substantial proportion of studies did not report participant age or gender, participation was male-skewed among studies that did report gender, and no non-binary participants were reported. Without adequate demographic reporting and subgroup-sensitive evaluation, it remains unclear whether reported safety and reliability findings generalise across the populations likely to interact with embodied AI systems. This confirms a broader concern in HRI research, namely that uneven gender representation and limited transparency about who is studied continue to weaken the interpretability of empirical findings [8, 34]. It also supports wider arguments that fairness in robotics is not a peripheral ethical issue, but a multidimensional concern spanning technical, social, and legal dimensions of deployment [17].

Within the reviewed corpus, fairness-related evidence was sparse and generally appeared only where bias, stereotyping, or differential treatment were treated as explicit objects of study. This is significant because fairness risks in embodied AI can emerge through several pathways. Physical safety and ergonomic suitability may not generalise across body types if interaction design is calibrated to an implicit default user [35]. Social harms may arise when robot behaviour reproduces or amplifies stereotypes [6, 5]. Demographic skew in training or evaluation data may undermine the reliability of perception and interaction models for underrepresented groups. The absence of non-binary participants also points to a broader identity-related risk, since embodied systems may reproduce binary assumptions in ways that generate exclusion or misrecognition for gender-diverse users [36].

Overall, these findings suggest that fairness in embodied AI should be treated as part of safety and trustworthiness rather than as a separate ethical layer. A minimum standard for future work should therefore include more consistent demographic reporting, more explicit attention to subgroup heterogeneity where relevant, and stronger justification where fairness-sensitive evaluation is not undertaken. Where systems rely on perception, prediction, or socially expressive behaviour, this should be complemented by intersectional auditing and targeted red-teaming of underrepresented user scenarios. More broadly, strengthening fairness in embodied AI will require not only technical improvements, but also changes in where, with whom, and under what conditions embodied systems

are evaluated. In this respect, more field-based and socially diverse studies will be important for addressing both the diversity gap and the realism gap identified in this review [34, 29, 37, 38, 39].

3.5 Conclusion

This chapter has examined how risk is studied in embodied HRI and what this evidence reveals about fairness in mixed-autonomy systems. Across the reviewed literature, risk assessment was dominated by controlled experimental designs, low ecological realism, and outcome-oriented measures focused mainly on safety and performance. By contrast, fairness-related risks, including bias, stereotyping, unequal treatment, and loss of agency, were far less frequently operationalised and measured.

The review also identified several limitations that are directly relevant to gender fairness. Participant reporting was often incomplete, demographic diversity was inconsistently documented, and non-binary representation was absent from the reviewed studies. In addition, dataset issues, including lab-only data and imbalance, were frequently underreported, which makes it difficult to assess whether model performance and interaction outcomes are robust across different user groups.

At the same time, the chapter shows that fairness-relevant signals are already present in the literature. Gender, age, digital literacy, workload, anxiety, and contextual factors all influenced how risks emerged and how embodied AI systems were experienced. However, these factors were more often treated as secondary findings than as central dimensions of trustworthy evaluation.

These results confirm the need for more systematic fairness-aware evaluation in embodied AI and HRI. In the context of SymAware, this means improving demographic reporting, making subgroup considerations more explicit, and operationalising fairness-related risks alongside safety and performance.

Gender Fairness in User Studies

This chapter describes how gender aspects were handled across the user studies conducted in SymAware. In line with Task 6.3, it focuses specifically on gender fairness in participant identification, recruitment, sample composition, study conduct, reporting, and interpretation. In this context, gender fairness is treated as both a methodological and an ethical issue.

As shown in the findings reported in Chapter 3, existing embodied AI, HRI, and MAS research often provides stronger evidence on safety and performance than on fairness, representativeness, and subgroup-sensitive interpretation. Incomplete demographic reporting and limited attention to underrepresented groups weaken the evidential basis for claims about equitable system behaviour. Against this background, the present chapter examines how gender was handled across SymAware's own user studies and what methodological lessons can be drawn from that experience.

The chapter discusses gender-fairness aspects in the following user studies:

A: Ronald Cumbal, Natalia Calvo-Barajas, Marina Escobar-Planas, Alexandros Rouchitsas, and Ginevra Castellano. 2025. *Visualizing Confidence in Delivery Robots: Insights from Two Online Studies.* In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (CHI EA '25). Association for Computing Machinery, New York, NY, USA, Article 578, 1–7.

B: Ronald Cumbal, Didem Gürdür Broo, and Ginevra Castellano. 2025. *Crowdsourcing eHMI Designs: A Participatory Approach to Autonomous Vehicle-Pedestrian Communication.* In *2025 34th IEEE International Conference on Robot and Human Interactive Communication* (RO-MAN), Eindhoven, Netherlands, 25–29 August 2025, 1320–1326.

C: Ronald Cumbal, Marcus Göransson, Alexandros Rouchitsas, Didem Gürdür Broo, and Ginevra Castellano. 2026. *A Collaborative Crowdsourcing Method for Designing External Interfaces for Autonomous Vehicles.* In Press, *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems* (CHI '26).

D: Alessio Galatolo, Ronald Cumbal, Alexandros Rouchitsas, Katie Winkle, Didem Gürdür Broo, and Ginevra Castellano. 2026. *Lightweight Visual Reasoning for Socially-Aware Robots.* Under review for *IEEE International Conference on Robotics and Automation* (ICRA).

E: Hong Wang, Natalia Calvo-Barajas, Katie Winkle, and Ginevra Castellano. 2025. *"Who Should I Believe?": User Interpretation and Decision-Making When a Family Healthcare Robot Contradicts Human Memory.* In *2025 34th IEEE International Conference on Robot and Human*

Interactive Communication (RO-MAN), Eindhoven, Netherlands, 25–29 August 2025, 1421–1428.

F: Marc Fraile, Natalia Calvo-Barajas, Anastasia Sophia Apeiron, Giovanna Varni, Joakim Lindblad, Nataša Sladoje, and Ginevra Castellano. 2025. *UpStory: The Uppsala storytelling dataset. Frontiers in Robotics and AI* 12, Article 1547578.

G: Mengyu Zhong, Lux Miranda, Fotios C. Papadopoulos, Katie Winkle, Alkistis Skalkidou, and Ginevra Castellano. 2026. *Designing Socially Assistive Robots for Perinatal Depression Screening: Insights and Ethical Considerations from Two Exploratory Studies. ACM Transactions on Human-Robot Interaction.*

H: Ana Tanevska, Katie Winkle, and Ginevra Castellano. 2025. *Information vs situation: balancing transparency and autonomy for trustworthy autonomous vehicles. Frontiers in Robotics and AI* 12, Article 1657857.

I: Ana Tanevska, Ananthapathmanabhan Ratheesh Kumar, Arabinda Ghosh, Ernesto Casablanca, Ginevra Castellano, and Sadegh Soudjani. 2025. *Blending Participatory Design and Artificial Awareness for Trustworthy Autonomous Vehicles.* In *2025 34th IEEE International Conference on Robot and Human Interactive Communication* (RO-MAN), Eindhoven, Netherlands, 25–29 August 2025, 156–162. IEEE.

J: Ana Tanevska, Arabinda Ghosh, Katie Winkle, Ginevra Castellano, and Sadegh Soudjani. 2023. *Communicating Awareness: Designing a Framework for Cognitive Human-Agent Interaction for Autonomous Vehicles.* Paper presented at *The Cars As Social Agents (CArSA): A Perspective Shift in Human–Vehicle Interaction Workshop*, 11th International Conference on Human-Agent Interaction (HAI 2023), 4 December 2023, Gothenburg, Sweden.

K: Sybert Stroeve, Ana Tanevska, Mirco Kroon, and Ginevra Castellano. *User evaluation by remote pilots of two types of detect-and-avoid systems: remain well clear bands versus route guidance.* In Aerospace MDPI (Under review).

4.1 Participant Identification and Recruitment

Across the project's user studies, gender was treated as a relevant dimension of participant diversity and reporting. This supported transparent sample description, assessment of representation, and reflection on whether findings might be shaped by cohort composition. As such, gender recording was not treated as an isolated measure, but as part of a broader fairness-sensitive approach. This included documenting inclusion and exclusion criteria, especially in studies involving targeted populations such as Studies F and G, making recruitment channels explicit, adapting study information to the participant group, and interpreting findings within the limits of the recruited sample.

The project user studies varied considerably in recruitment strategy and target population. Several online studies used Prolific, which made it possible to monitor gender composition and, in some cases, rely on platform-supported gender balancing, namely in Studies A, B, C, E, H, and I. Other studies relied on university-based (Study D), school-based (Study F), workshop-based (Study J), or expert recruitment (Study K), where gender balance was less controlled and more dependent on the available participant pool. Moreover, Study G, focused intentionally on

women with prior positive EPDS results in a perinatal mental-health context. These differences are important because fairness-sensitive interpretation depends not only on recording gender, but also on understanding how the recruitment model shapes inclusion and representation [8].

4.1.1 Gender composition across studies

Reported gender composition varied substantially across the project studies, reflecting differences in study format, recruitment strategy, and target population. Among the Prolific-based online studies, Study A included two samples with relatively balanced gender distributions, with 12 female and 14 male participants in the first study and 24 female and 32 male participants in the second. In this case, gender balance was supported automatically through the platform. Study B also achieved a balanced sample, comprising 38 female, 37 male, and 1 non-binary participant. Study C showed a similarly balanced composition, with 35 female, 31 male, and 1 non-binary participant. These studies therefore provide the clearest examples in the project of structured online recruitment supporting visible gender balance.

A similar pattern was observed in Study H, which included 104 female, 101 male, and 1 non-binary participant. Study I used the same participant dataset, as this dataset was sufficiently large and rich to support an additional analysis. As such, these studies stand out as the most balanced larger-scale samples in the project and also demonstrate that non-binary representation was included in some of the online work. Since Study I builds on the same participant composition as Study H, the importance of these studies lies in the extensive analysis of gender-related effects. By contrast, Study E, although also based on Prolific and using balancing procedures, resulted in a more male-skewed sample, with 70 female, 104 male, and 2 non-binary participants. This suggests that platform-based balancing can improve transparency and representation, but does not guarantee equal composition in every case.

The in-person studies showed greater variation. Study D, an in-person laboratory study recruited at the university, included 3 female, 6 male, and 1 non-binary participant. Although non-binary representation was present, the sample was small and more limited in representativeness. Study J, a workshop study, included 5 female, 6 male, and 4 non-binary participants. While the sample was small, it showed the strongest gender-diverse representation in proportional terms across the project. This is methodologically significant because it avoids reducing participant identity to a binary classification and supports a more inclusive account of who is represented in the study. From a gender-fairness perspective, the presence of non-binary participants improves the visibility of gender-diverse users in the evidence base, even if the study was not designed for detailed subgroup analysis [36]. In contrast, Study K, an expert online study with remote drone pilots, was strongly male-skewed, with 1 female and 8 male participants, which likely reflects the composition of the available expert participant pool.

The school-based study, Study F, involved 28 Year 2 pupils, equally divided between boys and girls, and 10 Year 3 pupils, with 4 boys and 6 girls. Although the cohort was small, the overall composition was relatively balanced. Study G differed from the other project studies because it consisted of two linked phases and focused intentionally on a narrowly defined participant group. Its participatory design phase involved seven women with previous positive EPDS results, and its evaluation phase involved five women who screened positive for EPDS. In this case, the key gender-fairness issue is not balance within the sample, but transparent acknowledgement that the study was intentionally restricted to women because the application focused on women with lived

experience of PND. This focus can be justified by the study aims, but it also means that the findings should be interpreted strictly within that scope, rather than generalised to perinatal users more broadly [11].

Overall, these studies show a mixed but informative pattern. Several online studies achieved relatively balanced gender composition and included non-binary participants, whereas locally recruited, workshop-based, expert, and targeted-population studies were more strongly shaped by the characteristics of the available participant pool or by the narrowness of the study scope. From a gender-fairness perspective, these differences matter because they affect not only representation, but also the strength and generalisability of the claims that can reasonably be made.

4.2 Practical Approaches to Gender Fairness

4.2.1 Recording and visibility of gender composition

Across the project, gender issues were handled in three main ways. First, gender was recorded and reported consistently enough to support transparency about participant composition. This is important because one of the problems identified in Chapter 3 was that gender reporting is often incomplete in the wider literature. In the SymAware studies, the explicit reporting of women, men, and, in several cases, non-binary participants made it possible to assess the degree of representation achieved in each study and to identify where important limitations remained.

4.2.2 Recruitment transparency and representational limits

Second, recruitment methods were selected and described in ways that made gender-related limitations visible. In the Prolific-based studies, platform recruitment supported a more structured approach to participant balancing and demographic monitoring. In the university-based, school-based, workshop-based, and expert studies, the recruitment context itself shaped the available participant pool more strongly. This did not necessarily imply unfair treatment, but it did affect representativeness. The project therefore treated recruitment source as part of the fairness picture rather than as a merely logistical detail.

4.2.3 Gender-sensitive design and substantive analysis

Third, some studies treated gender as more than a descriptive variable. On the one hand, Study I explicitly analysed sex-related differences in user behaviour through Markov-chain modelling. The analysis identified significant differences between male and female participants in transition patterns in both highway and suburban environments, and the homogeneity test further showed that the sex subgroups did not share the same underlying transition matrix. This is important from a gender-fairness perspective because it shows that participant characteristics can affect interaction dynamics in substantive ways rather than functioning only as background descriptors. At the same time, the analysis relied on binary sex categories and therefore did not capture the broader gender diversity represented elsewhere in the project. Study H used the same participant pool and dataset as Study I, rather than collecting additional data, and provided a complementary analysis of gender-related effects. The results described in Study H show that gender had a weak negative correlation with participants' perceptions and attitudes towards AV and a weak positive correlation with trust ratings. These findings suggest that gender-related effects may be present even when

they are modest in strength. As such, these studies suggest that gender can be analytically relevant in different ways, ranging from weak attitudinal associations to stronger differences in behavioural transition patterns.

On the other hand, Study G addressed gender issues not only through participant selection, but also through design. Across its two linked phases, participants reflected on how robot identity cues such as voice gender, facial appearance, ethnicity, age, and anthropomorphic style affected comfort, trust, and willingness to engage in a perinatal mental-health screening context. Participants emphasised that being able to choose a gendered voice could increase comfort and acceptability, especially in a sensitive healthcare setting, and some described strong preferences linked to previous healthcare experiences. At the same time, the study showed that asking users to personalise multiple identity cues directly could add to the already high emotional and cognitive burden of the screening context. These findings informed a design approach based on a relatively neutral default robot appearance with limited gender and ethnic cues, combined with optional personalisation of selected features such as voice and face. From a gender-fairness perspective, this is important because it shows that gender issues in human-agent interaction do not arise only through participant composition, but also through the design of identity cues and the extent to which users can feel recognised, comfortable, and emotionally safe without being overburdened by the interaction [10].

4.2.4 Interpretation in light of study scope

Finally, the findings of the studies were interpreted in light of their actual gender composition and study scope. This was is especially visible in three cases. The first is Study G, where the women-only design was integral to the target population and study purpose [40]. The second is Study K, where the expert sample was strongly male-dominated and therefore offers limited evidence about gender-diverse user response. The third is Study J, where a relatively small workshop sample nevertheless achieved stronger non-binary representation than any of the larger online studies [41]. Together, these examples show that gender fairness is not reducible to sample size alone. It depends on the relation between study scope, participant access, design choices, and how the resulting evidence is interpreted.

4.3 Summary of Implemented Approaches to Gender Fairness

Across the SymAware studies, gender fairness was addressed through four main strategic approaches. First, gender was systematically recorded and reported, making participant composition visible and supporting fairness-sensitive interpretation. Compared with the broader evidence base reviewed in Chapter 3, this represents a significant strength, since incomplete demographic reporting remains a persistent limitation in embodied AI and HRI research.

Second, different recruitment strategies were used to support balanced participation. This was most effective in the structured online studies, especially Studies A, B, C, H, and I, where platform-based tools enabled closer monitoring of gender composition. Notably, several studies, namely Studies B, C, D, E, H, I, and J, included non-binary participants, thereby moving the project beyond a strictly binary reporting framework.

Third, in some studies gender was treated as an analytical or design-relevant factor rather than merely as a descriptive variable. Study I identified sex-related differences in behavioural transition patterns, Study H found statistically significant associations between gender, trust ratings, and

AV-NARS scores, and Study G addressed gender as a design issue linked to identity and emotional safety in a healthcare context.

Fourth, the project placed emphasis on interpretative transparency. Because the studies varied substantially in scale and scope, ranging from broad online samples to narrow expert pools, fairness was approached not as a rigid threshold, but as a matter of making scope explicit and exercising caution in generalisation. Despite these strengths, important limitations remained. Descriptive reporting was generally stronger than subgroup-level inference, and in some cases, most notably Study I, the analysis relied on binary sex categories due to data constraints. Furthermore, although the inclusion of non-binary participants was an important strength of the project, this did not always translate into dedicated analytical treatment of gender-diverse experiences.

4.4 Conclusion

This chapter has detailed how gender fairness was addressed across the SymAware user studies. By treating gender as a relevant dimension of participant diversity and study reporting, the project achieved a level of transparency that exceeds what is often visible in the wider HRI literature [8].

The project also shows that gender fairness is multifaceted. It concerns not only recruitment and sample composition, but also analytical rigour, such as the identification of sex-related behavioural differences, and design choices, including the handling of robot identity cues. These efforts strengthen both the ethical grounding and the methodological robustness of the project's findings.

Ultimately, these results align with the *Gendered Innovations 2* framework, which argues that integrating sex and gender analysis increases the societal relevance and research quality of scientific and technological work [11]. Within SymAware, this approach has helped position gender fairness not as a procedural checkbox, but as a necessary condition for producing trustworthy knowledge about human–agent interaction and multi-agent systems.

Bibliography

[1] M. R. Endsley, "Toward a theory of situation awareness in dynamic systems," *Human factors*, vol. 37, no. 1, pp. 32–64, 1995.

[2] E. Commission, "Ethics guidelines for trustworthy ai," 2019, accessed: 2024-06-05. [Online]. Available: https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai

[3] J. Schoeffer, M. De-Arteaga, and N. Kuehl, "Explanations, fairness, and appropriate reliance in human-ai decision-making," in *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, 2024, pp. 1–18.

[4] J. J. Aloor, S. N. Nayak, S. Dolan, and H. Balakrishnan, "Cooperation and fairness in multi-agent reinforcement learning," *Journal on Autonomous Transportation Systems*, vol. 2, no. 2, pp. 1–25, 2024.

[5] M. T. Parreira, S. Gillet, and K. Winkle, "How did we miss this? a case study on unintended biases in robot social behavior," 2023. [Online]. Available: https://doi.org/10.1145/3568294.

[6] T. Hitron, N. Morag, and H. Erel, "Implications of ai bias in hri: Risks (and opportunities) when interacting with a biased robot," in *ACM/IEEE International Conference on Human-Robot Interaction*. IEEE Computer Society, 3 2023, pp. 83–92.

[7] A. Habibovic, V. M. Lundgren, J. Andersson, M. Klingegård, T. Lagström, A. Sirkka, J. Fagerlönn, C. Edgren, R. Fredriksson, S. Krupenia *et al.*, "Communicating intent of automated vehicles to pedestrians," *Frontiers in psychology*, vol. 9, p. 1336, 2018.

[8] K. Winkle, E. Lagerstedt, I. Torre, and A. Offenwanger, "15 years of (who) man robot interaction: Reviewing the h in human-robot interaction," *ACM Transactions on Human-Robot Interaction*, vol. 12, no. 3, pp. 1–28, 2023.

[9] P. Hall and D. Ellis, "A systematic review of socio-technical gender bias in ai algorithms," *Online Information Review*, vol. 47, no. 7, pp. 1264–1279, 2023.

[10] L. Tanqueray, T. Paulsson, M. Zhong, S. Larsson, and G. Castellano, "Gender fairness in social robotics: exploring a future care of peripartum depression," in *2022 17th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE, 2022, pp. 598–607.

[11] E. Commission, D.-G. for Research, and Innovation, *Gendered innovations 2 â€" How inclusive analysis contributes to research and innovation â€" Policy review*. Publications Office of the European Union, 2020.

[12] I. M. Penteado, A. C. S. do Nascimento, D. Corrêa, E. A. F. Moura, R. Zilles, M. C. R. L. Gomes, F. J. Pires, O. S. Brito, J. F. da Silva, A. V. Reis *et al.*, "Among people and artifacts: Actor-network theory and the adoption of solar ice machines in the brazilian amazon," *Energy research & social science*, vol. 53, pp. 1–9, 2019.

[13] N. Cila and C. DiSalvo, "What can actor-network theory teach us about the socio-technological implications of delivery robots?" in *Designing smart objects in everyday life: Intelligence, Agencies, Ecologies*. Bloomsbury, 2021, pp. 109–124.

[14] N. Calvo-Barajas, A. Rouchitsas, and D. Broo, "Examining human-robot interactions: Design guidelines for trust and acceptance," *Interdisciplinary Dimensions of Human-Technology Interaction, Springer*, In press.

[15] High-Level Expert Group on Artificial Intelligence, "Ethics guidelines for trustworthy ai," 2019, accessed: 2025-03-21. [Online]. Available: https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai

[16] E. A. I. Act, "The eu artificial intelligence act," *European Union*, 2024.

[17] E. Fosch-Villaronga, A. Mut-Piña, M. R. Shaffique, and M. Schwed-Shenker, "Exploring fairness in service robotics," in *Cambridge Forum on AI: Law and Governance*, vol. 1. Cambridge University Press, 2025, p. e28.

[18] H. Detjen, S. Faltaous, B. Pfleging, S. Geisler, and S. Schneegass, "How to increase automated vehiclesâ€™ acceptance through in-vehicle interaction design: A review," *International Journal of Human–Computer Interaction*, vol. 37, no. 4, pp. 308–330, 2021.

[19] J. Kraus, F. Babel, P. Hock, K. Hauber, and M. Baumann, "The trustworthy and acceptable hri checklist (ta-hri): questions and design recommendations to support a trust-worthy and acceptable design of human-robot interaction," *Gruppe. Interaktion. Organisation. Zeitschrift Für Angewandte Organisationspsychologie (GIO)*, vol. 53, no. 3, pp. 307–328, 2022.

[20] T. Reinmund, P. Salvini, L. Kunze, M. Jirotka, and A. F. Winfield, "Variable autonomy through responsible robotics: Design guidelines and research agenda," *ACM Transactions on Human-Robot Interaction*, 2023.

[21] M. D. Peters, C. Godfrey, P. McInerney, Z. Munn, A. C. Tricco, and H. Khalil, "10. scoping reviews," *JBI manual for evidence synthesis*, pp. 417–476, 2024.

[22] M. Barreda-Ángeles, S. Amendola, T. Hartmann, A. Meier, K. Mansfield, S. Coecke, and I. Hupont, "Virtual worlds and mental health: A scoping review," 2025.

[23] G. Abercrombie, D. Benbouzid, P. Giudici, D. Golpayegani, J. Hernandez, P. Noro, H. Pandit, E. Paraschou, C. Pownall, J. Prajapati *et al.*, "A collaborative, human-centred taxonomy of ai, algorithmic, and automation harms," *arXiv preprint arXiv:2407.01294*, 2024.

[24] K. E. Schaefer, J. Y. Chen, and J. L. Szalma, "A meta-analysis of factors influencing the development of trust in automation: Implications for understanding autonomy in future systems," *Human factors*, vol. 58, no. 3, pp. 377–400, 2016.

[25] G. Angelopoulos, L. Mangiacapra, A. Rossi, C. D. Napoli, and S. Rossi, “What is behind the curtain? increasing transparency in reinforcement learning with human preferences and explanations,” *Engineering Applications of Artificial Intelligence*, vol. 149, 6 2025.

[26] G. Campagna, M. Lagomarsino, M. Lorenzini, D. Chrysostomou, M. Rehm, and A. Ajoudani, “Estimating trust in human-robot collaboration through behavioral indicators and explainability,” *IEEE Robotics and Automation Letters*, vol. 10, pp. 10 218–10 225, 2025.

[27] D. Kontogiorgos and J. Shah, “Questioning the robot: Using human non-verbal cues to estimate the need for explanations,” in *ACM/IEEE International Conference on Human-Robot Interaction*. IEEE Computer Society, 2025, pp. 717–728.

[28] M. Natarajan and M. Gombolay, “Trust and dependence on robotic decision support,” *IEEE Transactions on Robotics*, vol. 40, pp. 4670–4689, 2024.

[29] O. E. K. Lee, J. C. Yun, and D. H. Park, “Perceptions and experiences of korean american older adults with companion robots through long-term use: a comparative analysis of robot retention vs. return,” *Frontiers in Public Health*, vol. 12, 2024.

[30] A. Arif, Z. Zakeri, A. Omurtag, P. Breedon, and A. Khalid, “Neurophysiological approach for psychological safety: Enhancing mental health in humanâ€“robot collaboration in smart manufacturing setups using neuroimaging,” *Information (Switzerland)*, vol. 15, 10 2024.

[31] A. Villalonga, Y. Cruz, R. Haber, L. Bastida, S. Sillaurren, and F. Castaño, “Edge-computing framework for human-robot collaboration in industry 5.0: Enhancing operator well-being and efficiency in manufacturing,” 2024.

[32] G. Campagna, D. Chrysostomou, and M. Rehm, “Analysis of facial features for trust evaluation in industrial human-robot collaboration,” in *Proceedings of IEEE Workshop on Advanced Robotics and its Social Impacts, ARSO*. IEEE Computer Society, 2024, pp. 1–6.

[33] T. Whiting, A. Gautam, J. Tye, M. Simmons, J. Henstrom, M. Oudah, and J. W. Crandall, “Confronting barriers to human-robot cooperation: balancing efficiency and risk in machine behavior,” *iScience*, vol. 24, 1 2021.

[34] K. Seaborn, G. Barbareschi, and S. Chandra, “Not only weird but "uncanny"? a systematic review of diversity in human–robot interaction research,” *International Journal of Social Robotics*, vol. 15, no. 11, pp. 1841–1870, 2023.

[35] S. Kaur, M. Y. Chan, N. Islan, J. Oyekan, and A. Tiwari, “Applying machine learning models to the development of ergonomic human-robot work envelopes,” in *Proceedings of IEEE Workshop on Advanced Robotics and its Social Impacts, ARSO*. IEEE Computer Society, 2025, pp. 93–99.

[36] M. Stolp-Smith and T. Williams, “More than binary: Transgender and non-binary perspectives on human robot interaction,” in *Proceedings of the 2024 ACM/IEEE international conference on human-robot interaction*, 2024, pp. 697–705.

[37] H. Chen, Y. Kim, K. Patterson, C. Breazeal, and H. W. Park, "Social robots as conversational catalysts: Enhancing long-term human-human interaction at home," *Science Robotics*, vol. 10, no. 100, p. eadk3307, 2025.

[38] H.-R. Ho, E. M. Hubbard, and B. Mutlu, "" it's not a replacement:" enabling parent-robot collaboration to support in-home learning experiences of young children," in *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, 2024, pp. 1–18.

[39] J. Mišeikis, P. Caroni, P. Duchamp, A. Gasser, R. Marko, N. Mišeikienė, F. Zwilling, C. De Castelbajac, L. Eicher, M. Früh *et al.*, "Lio-a personal robot assistant for human-robot interaction and care applications," *IEEE robotics and automation letters*, vol. 5, no. 4, pp. 5339–5346, 2020.

[40] M. Zhong, L. Miranda, F. C. Papadopoulos, K. Winkle, A. Skalkidou, and G. Castellano, "Designing socially assistive robots for perinatal depression screening: Insights and ethical considerations from two exploratory studies," *J. Hum.-Robot Interact.*, Feb. 2026, just Accepted. [Online]. Available: https://doi.org/10.1145/3797257

[41] A. Tanevska, A. Ghosh, K. Winkle, G. Castellano, and S. Soudjani, "Communicating awareness: Designing a framework for cognitive human-agent interaction for autonomous vehicles," in *Cars As Social Agents (CarSA): A Perspective Shift in Human-Vehicle Interaction.*, 2023.